COMBINATORICS
ASSIGNMENT 5

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1. Problem 1

In how many ways can we build a 7-beaded circular necklace if there are 5 available types of gemstones (repetition) allowed and all rotations and reflections of a given necklace are equivalent. What if we have q gemstones?

Solution. Our goal is an application of Burnside's Lemma here. Now, for Burnside's Lemma, we need to start with a finite group G and a set X and G acting on X. For each $g \in G$, set X^g to be the elements of X that are fixed by g's action on X. That is to say we set $X^g = \{x \in X : g \cdot x = x\}$. Burnside's Lemma then asserts that the number of G orbits of X, denoted $|G/X|$ is given by

$$|G/X| = \frac{1}{|G|} \sum_{g \in G} |X^g|.$$

We start by noticing that in our problem, we are dealing with the group of rotations and reflections on the 7-cycle, which is commonly referred to as the dihedral group on 7 elements, D_7. Notice that the size of $|D_7| = 14$.

Assume we have q gemstones for the sake of generality. Now, we must account for all of the elements of D_7 in order to account for the summation in Burnside's Lemma.

So we begin with the identity rotation which fixes our 7 places with q colorings counting q^7 occurrences. We then proceed to counting the non-trivial rotations of which there are 6, so we get another $6q$ coloring occurrences there. Lastly, we consider the reflections, for which there are 7 axis of symmetry each bead on the axis can be any of q colors and the remaining 6 beads are set up in 3 pairs where each of the 3 pairs must match in color. Thus, we count this number of colorings as $7 \cdot q \cdot q^3 = 7q^4$.

So when we put all this together we see that

$$|D_7/[q]| = \frac{1}{14}(q^7 + 6q + 7q^4).$$

So for 5 colors we have

$$|D_7/[5]| = \frac{1}{14}(5^7 + 6 \cdot 5 + 7 \cdot 5^4) = 5895$$

$\square$

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2. Problem 2

A rectangular prism has equilateral triangles for its base and top, and three rectangular lateral faces. In how many ways can we color if the faces can be colored using any one of the 5 colors.

Solution. We again appeal to Burnside's Lemma. We begin by determining the rotational symmetry group of the prism, which is the identity element, a rotation about the axis through the triangular faces of which there are 3 $120°$, and a rotation about an axis taken through one of the rectangular faces, a $180°$ rotation which makes the group of rotations of size 6.

Now to the summation portion of Burnside's Lemma. For the identity element coloring the faces with 5 colors, because there are 5 faces, we have 5^5 possibilities.

For the rotations about the axis through the triangular faces, for this to remain the same after a rotation we must have any of the 5 colors on either end of the prism in the triangular face. But, for the rectangular faces, those must all be the same color. So we have 5^3 possibilities here of which there are 3 rotations.

Now what if we put the axis of rotation through a rectangular face? For the prism to remain unchanged through the rotation the face can be chosen to be any of our 5 colors, the other two rectangles must be the same color, and as well the triangles must maintain the same color. So that is 3 regions with 5 colors to choose from which yields 5^3 possibilities of which there are 2 rotations.

To add everything up we see that

$$\# = \frac{1}{6}(5^5 + 3(5^3) + 2(5^3)) = 625.$$

□

3. Problem 3

Find the pattern inventory of the red-blue vertex colorings of a tetrahedron. Repeat the problem with red-blue edge colorings.

Solution. □

4. Problem 4

How many 10 red, 12 blue, 3 green squares are there in the coloring of a 5×5 chessboard?

Solution. □

5. Problem 5

Find the pattern inventory of the red-blue vertex colorings of a floating cube (i.e. the cube can freely move in space).

Solution. □